

SUBIN GHIMIRE

Portfolio: itzsubin.vercel.app

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Education

University of North Texas

Bachelor of Science in Computer Science, Honors College

Denton, TX

Expected: Dec 2027

Skills

Programming: C/C++, Python, JavaScript, TypeScript, Dart, Tailwind CSS

Frameworks/Libraries: React, Node.js, Express.js, Vite, Hono, Flutter, Framer Motion

Databases: PostgreSQL, MongoDB, VectorDB

Tools & Technologies: Docker, Git, GitHub, REST APIs, NumPy, Pandas, Linux, Google Maps API

Experience

IT Support

May 2025 – Present

University of North Texas

Denton, TX

- Resolved complex hardware, software, and network issues for housing and dining divisions utilizing command-line tools and system diagnostics across Windows and Linux environments.
- Managed, tracked, and documented technical incidents within ServiceNow, maintaining high SLA compliance and resolution efficiency.

Undergraduate Research Assistant

October 2025 – Present

University of North Texas

Denton, TX

- Engineered high-throughput data preprocessing pipelines using NumPy and Pandas to clean, transform, and normalize raw spatial tracking datasets under Prof. Heng Fan.
- Developed and applied mathematical models to calculate frame-by-frame coordinate transformations, accelerating video-based tracking data preparation for downstream analysis models.

Projects

Allerly | *Flutter, Dart, Next.js, TypeScript, Supabase, PostgreSQL, Google Maps API, Google ML Kit*

- Engineered a cross-platform mobile restaurant discovery application featuring dynamic location-based menu auditing using the Google Maps API and Geolocator.
- Integrated Google ML Kit Text Recognition to build an on-device camera pipeline, extracting text from physical menus and parsing it instantly against a relational database.
- Deployed scalable serverless Supabase Edge Functions in TypeScript to handle real-time cross-referencing of menu structures against user allergy profiles.

StudyPal | *React, TypeScript, Tailwind CSS, PostgreSQL, Hono, OpenRouter APIs, Cloudflare*

- Built a full-stack AI-powered study platform implementing intelligent note-taking, adaptive quiz generation, and active recall flashcards.
- Implemented the SM-2 spaced-repetition algorithm using a PostgreSQL backend to dynamically calculate optimal card review intervals, maximizing user retention metrics.

Airline Route Optimizer | *C++*

- Designed and implemented an optimized multi-city routing graph application to compute efficient flight paths and ideal central meeting points.
- Utilized memory-efficient custom adjacency lists and implemented Breadth-First Search (BFS) pathfinding to minimize computational complexity during route discovery.

Leadership & Activities

Nepalese Student Association | *UNT*

January 2025 - Present

Undergraduate Representative

- Direct onboarding initiatives and transitional support systems for international and domestic students, coordinating academic and cultural integration workshops.